

Surnames and Name _____

MARK _____ points/10

Important: Please, write your name and last name before you start. Neither books nor any other extra material are allowed in the exam but the calculator. Please write down your answers on the exam sheet. No additional or extra sheets will be corrected. You may use the back of the sheet if needed. In the test part of the exam mark with an X the right answer, no crossings out and no additional comments are allowed. Exam will take **50 minutes**. Please, place your ID card or your student card on your table.

1. (5 points) Mark with an X the right answer, **there is only one correct answer per question**. (Correct answers mark as +0.5, incorrect ones mark as - 0.15, and non answered questions mark as 0. The resulting mark will not be smaller than 0 in any case).

4,35 ✓

1	Which one of the following statements is FALSE :
X	Polarity of a molecule depends exclusively on the polarity of the chemical bonds.
	Ionic compounds do not conduct electricity except in solution and molten state.
	Hydrogen acquires a partial positive charge when it bonds with an electronegative atom.
	Materials that present strong atomic bonding have low linear coefficients of thermal expansion.

✓

2	A solid substance was tested in the laboratory and the results are listed below. Identify the substance based on them.
	• It dissolves in water • It is an electrolyte • It melts at a high temperature
	Cu
X	CuBr ₂
	Graphite
	C ₆ H ₁₂ O ₆ (sugar)

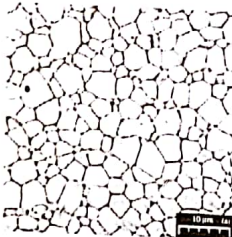
✓

3	The plane (1 $\bar{1}$ 1) is parallel to:
	(1 1 1)
	($\bar{1}$ $\bar{1}$ 1)
X	($\bar{1}$ 1 $\bar{1}$)
	($\bar{1}$ 1 1)

4	Determine the Miller indices of the line of intersection of $(\bar{1}\bar{1}1)$ and (110) :
☐	$[110]$
☐	$[101]$
☐	$[10\bar{1}]$
☒	$[\bar{1}10]$

5	Schottky defect in ionic crystals is a:
☐	Interstitial impurity
☒	Pair of nearby cation and anion vacancies
☐	Vacancy-interstitial pair of cations
☐	Substitutional impurity

6	Calculate the activation energy for vacancy formation in aluminum given that the equilibrium number of vacancies at 500 °C is 7.55×10^{23} vacancies/m ³ . The atomic weight and density (at 500 °C) for aluminum are, respectively, 26.98 g/mol and 2.62 g/cm ³ . $K = 8.62 \times 10^{-5}$ eV/K; $N_A = 6.022 \times 10^{23}$ atoms/mol.
☐	1.55 eV/atom
☐	2.63 eV/atom
☒	0.75 eV/atom
☐	1.29 eV/atom

7	The figure in an ionic solid is a(n):
	
☒	Grain boundary
☐	Edge dislocation
☐	Frenkel defect
☐	Twin boundary

8	Which of these variables do not affect diffusion for a case that is described by Fick's First Law?
☐	Temperature
☐	Concentration
☒	Number of vacancies
☒	Time

9	Diffusivity is greater when...
X	solute diffuses interstitially
	temperature is decreased
	crystalline structure is more compact
	number of vacancies is decreased

10	Determine the number of kilograms of hydrogen that pass per hour through a 5 mm thick sheet of palladium having an area of 0.25 m ² at 500 °C. Assume a diffusion coefficient of 1.0×10^{-8} m ² /s, that the concentrations at the high- and low-pressure sides of the plate are 2.4 and 0.6 kg of hydrogen per cubic meter of palladium, and that steady-state conditions have been attained.
X	3.2×10^{-3} kg/h
	0.9×10^{-6} kg/h
	3.2×10^{-6} kg/h
	6.8×10^{-3} kg/h

4,75

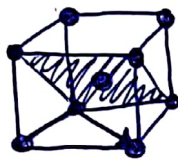
2. (5 points) The plane $(1\ 0\ \bar{1})$ of a cubic cell is shown below; the circles represent atoms and they correspond to iron. Atomic linear density for the direction $[1\ 1\ 0]$ is 2.46 atoms/nm.



- a) Indicate the type of crystalline structure, the lattice parameter (a) and the density of the metal in g/cm^3 .
 b) Calculate the planar density in the plane $(1\ 1\ 0)$ in atoms/cm^2 .
 c) After a certain thermal treatment, density of iron results $7.83\ \text{g/cm}^3$. Determine the total number of vacancies in a cubic centimeter of Fe that appear after the increase in temperature.
 d) Calculate the Burgers vector magnitude for the dislocations movement in the more favourable direction for the Fe at room temperature.

Data: $M(\text{Fe}) = 55.8\ \text{g/mol}$; $N_A = 6.022 \times 10^{23}\ \text{atoms/mol}$.

a)



Body centered cubic ✓

$$\rho_{[110]} = \frac{1}{L} = 2.46\ \text{atoms/nm} \quad L = 0.4065\ \text{nm}$$



$$a = \frac{4R}{\sqrt{3}}$$

$$\sqrt{2}a = L$$

$$a = 0.2874\ \text{nm}$$

$$\rho_v = \frac{M \cdot n}{N_A \cdot V}$$

$$= \frac{55.8\ \text{g/mol} \cdot 2}{6.022 \cdot 10^{23}\ \text{atoms/mol} \cdot (0.2874 \cdot 10^{-7}\ \text{cm})^3} = 7.81\ \text{g/cm}^3 \quad \checkmark$$

1,25

b)

$$\rho_{(110)} = \frac{2}{\sqrt{2}a \cdot a}$$

$$= \frac{2}{\sqrt{2} \cdot 0.2874 \cdot 10^{-7}\ \text{cm} \cdot 0.2874 \cdot 10^{-7}\ \text{cm}} = 1.71 \cdot 10^{15}\ \text{atoms/cm}^2 \sim 1.71\ \text{atoms/nm}^2 \quad 1,25$$



c)

$$7.81\ \text{g/cm}^3 \rightarrow 7.83\ \text{g/cm}^3$$

1,0

$$7.83\ \text{g/cm}^3 = \frac{55.8(1-x)\ \text{g/mol} \cdot \text{at}}{6.022 \cdot 10^{23}\ \text{atoms/mol} \cdot (0.2874 \cdot 10^{-7}\ \text{cm})^3}$$

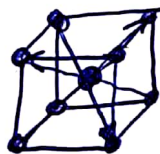
$$x = -0.006 \quad ??$$

Vacancies per unit cell

$$0.006\ \text{vac/cell} \cdot \frac{1\ \text{cell}}{a^3\ \text{cm}^3} =$$

$$= 2.53 \cdot 10^{20}\ \text{vac/cm}^3$$

d)



$\langle 111 \rangle$

$$D = 4R = 2|\bar{b}|$$

$$|\bar{b}| = 2R = \frac{\sqrt{3}a}{2} = 0.2489\ \text{nm} \quad 1,25$$

$$R = \frac{\sqrt{3}a}{4}$$

MORE COMPACT
DIRECTION
(CUBE DIAGONALS)